

EXPERIMENT 6 - SMART SURVEILLANCE SYSTEM USING ESP32-CAM

MCTA 3203 - MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

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Abstract

This experiment focuses on developing a simple smart surveillance system by integrating the ESP32-CAM module with a motorized servo platform and manual input control. The system is designed to stream live video over Wi-Fi while enabling horizontal camera panning using a servo motor controlled through PWM signals. A push button was incorporated to replace continuous panning with manual activation, allowing the camera to move only when the button is pressed. The experiment demonstrates the integration of key mechatronic elements including wireless communication, camera interfacing, actuator control, and digital input handling. Through hardware setup, coding, and testing, the prototype successfully streamed video, responded to button-triggered movement, and showcased how IoT-based surveillance can be implemented using microcontrollers. The lab highlights the practical application of mechatronic system integration and provides foundational understanding for more advanced surveillance features such as motion detection or face recognition.

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Introduction

TASK 1 - Camera ,Servo and Push Button Integration

The first task focuses on enhancing the basic ESP32-CAM smart surveillance system by adding a manual input component that allows the user to control the horizontal panning of the camera. In the original setup, the servo motor rotates continuously between its minimum and maximum angles, simulating automatic environmental scanning. While this provides wide-area coverage, it lacks user flexibility and direct interaction. Therefore, the system is modified so that the servo only moves when a push button is pressed, giving users intentional control over camera direction.

To achieve this functionality, the ESP32-CAM module plays a central role by managing video streaming, reading digital input states, and generating PWM signals for the servo motor through the LEDC timer. The integration of the push button introduces concepts such as internal pull-up resistors, digital signal readings, and debouncing considerations. These elements ensure that the servo responds accurately and reliably only when the button is activated. The task reinforces essential mechatronics concepts, including sensor–actuator integration, embedded software development, and user-activated control.

This manual control approach provides students with a richer understanding of interactive surveillance systems. By modifying the system to respond to user input instead of automated continuous movement, the surveillance prototype becomes more versatile and practical. The task demonstrates how simple input devices can influence actuator behaviour and highlights the importance of designing systems that accommodate both automation and manual operation.

TASK 2 - Face Detection

The second task expands the capabilities of the ESP32-CAM surveillance system by enabling its built-in face detection features. Unlike the first task, which focuses on manual actuation, this task introduces autonomous intelligence into the system through embedded computer vision. The ESP32-CAM utilizes the ESP-WHO framework to process live video frames and detect human faces in real time. This allows the device to identify facial structures and draw bounding boxes around detected faces before transmitting the processed video stream.

Implementing face detection exposes students to fundamental concepts in machine vision, such as feature extraction, classification, and frame buffer manipulation. Despite the hardware limitations of microcontrollers, the ESP32-CAM is capable of running optimized Haar cascade algorithms that allow it to detect faces with high efficiency. This embedded AI processing demonstrates how modern surveillance systems can operate independently of external computers or cloud-based services.

The addition of face detection significantly increases the functionality and potential applications of the surveillance prototype. It transforms the ESP32-CAM from a simple video streaming device into an intelligent monitoring platform capable of recognizing human presence. This feature also lays the foundation for future enhancements such as automatic servo tracking, motion-triggered alerts, or identity-based recognition. Through this task, students gain practical experience with integrating artificial intelligence into IoT systems and understand how computer vision contributes to smart security technologies.

Material & Equipment

TASK 1 - Camera ,Servo and Push Button Integration

The following components and equipment were used to complete Task 1:

(From the official lab sheet + required additions for Task 1)

Hardware

- ESP32-CAM Module (AI-Thinker, OV2640 camera)
- SG90 / MG90S Servo Motor
- Pushbutton switch (1 unit)
- 10k Ω resistor (optional – for external pull-up)
- Breadboard
- Jumper wires
- USB-to-Serial Programmer (FTDI or CH340)
- 5V 2A power supply (recommended for stable servo operation)
- Computer with Arduino IDE installed

Software

- Arduino IDE
- ESP32 board package
- ESP32-CAM CameraWebServer base code (modified)

TASK 2 - Face Detection

Hardware

- ESP32-CAM Module with OV2640 camera

- SG90 / MG90S servo motor (optional for face-triggered motion)
- USB-to-Serial programmer (FTDI/CH340)
- Jumper wires
- 5V power supply
- Laptop with Arduino IDE

Software

- Arduino IDE + ESP32 board
- ESP32-CAM face detection library (included in example: CameraWebServer)
- Modified version of the “CameraWebServer” code with face detection enabled

Experimental Setup

TASK 1 - Camera ,Servo and Push Button Integration

ESP32-CAM Programming Setup (Using Micro-USB Slot)

- Connect ESP32-CAM to laptop using micro-USB cable.
- No FTDI or CH340 is needed.
- ESP32-CAM auto-enters upload mode when uploading through USB.
- Select correct COM port in Arduino IDE.

Servo Motor Setup

- Servo signal wire connected to GPIO 2.

- Servo VCC wire connected to 5V.
- Servo GND wire connected to GND.
- Camera mounted on the servo base for horizontal panning.

Pushbutton Setup

- One leg of pushbutton connected to GPIO 13.
- Other leg connected to GND.
- GPIO 13 configured with internal pull-up.

System Arrangement

- ESP32-CAM powered through external 5V for stable camera + servo operation.
- Camera streams video through Wi-Fi.
- Servo moves only when button is pressed.

TASK 2 - Face Detection

3.1 ESP32-CAM Initialization

- ESP32-CAM is configured using the standard pin mapping from the lab sheet.
- Pixel format: JPEG
- Frame size: VGA
- Camera frequency: 20 MHz

3.2 Enabling Face Detection

The face detection module relies on the internal ESP-WHO library.
In the CameraWebServer example, two parameters are important:

```
s->set_framesize(s, FRAMESIZE_QVGA);
```

```
enable_detection = true;
```

3.3 Hardware Configuration (Optional Servo Trigger)

If servo triggers on face detection:

- **Servo Signal** → **GPIO 2** on ESP32-CAM
- **Servo VCC** → **5V** on ESP32-CAM
- **Servo GND** → **GND** on ESP32-CAM

3.4 System Layout

- ESP32-CAM placed facing user
- Connect to WiFi
- Access web interface → face detection overlay appears (green square)
- Optional: when face detected → servo rotates or image captured

Methodology

TASK 1 - Camera ,Servo and Push Button Integration

1. Software Preparation

1. Install Arduino IDE.
2. Install the ESP32 board package via Arduino IDE Board Manager.

3. Select “ESP32 Wrover Module” as the board.
4. Set Partition Scheme to “Huge App (3MB No OTA)”.
5. Set Upload Speed to 115200.
6. Connect the ESP32-CAM via micro-USB.
7. Select the correct COM port in Arduino IDE.

2. Hardware Integration

Button Integration

1. Connect a push button to GPIO 14.
2. Configure the pin in code:

```
pinMode(button_pin, INPUT_PULLUP);
```

3. Detect button press:

```
bool button_pressed = digitalRead(button_pin) == LOW; // LOW = pressed
```

4. Button released = HIGH.

Servo Motor Integration

1. Connect a **servo motor** to **GPIO 2**.
2. Configure PWM for servo:

```
ledcSetup(0, 50, 16);          // channel 0, 50Hz, 16-bit resolution
```

```
ledcAttachPin(servo_pin, 0);
```

```
ledcWrite(0, servo_position);
```

3. Servo Movement Logic

1. When the button is pressed, the servo moves between its minimum and maximum PWM positions.
2. Direction changes automatically when a limit is reached.
3. When the button is released, the servo holds its last position.

Servo sweep code snippet:

```
if (button_pressed) {  
    servo_position += servo_speed;  
    if (servo_position > servo_max_position) servo_position =  
servo_max_position;  
} else {  
    servo_position -= servo_speed;  
    if (servo_position < servo_min_position) servo_position =  
servo_min_position;  
}  
ledcWrite(0, servo_position);
```

4. ESP32-CAM Video Streaming

1. Connect the ESP32-CAM to WiFi:

```
WiFi.begin(ssid, password);  
while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
}
```

2. Initialize the AI-Thinker camera module: configure pins, frame size, and JPEG quality.
3. Start the HTTP server to serve:
 - Web page (/) showing camera feed and stream link.
 - MJPEG stream (/stream) for live video.

4. The servo movement runs simultaneously in the streaming loop.

5. Flow Summary

1. Power on ESP32-CAM and connect to WiFi.
2. Access the camera web page via browser.
3. Press the button to move the servo back and forth.
4. Release the button to stop the servo.
5. Observe the live video stream while the servo operates.

6. Full Code

```
#include "esp_camera.h"

#include <WiFi.h>

#include "esp_http_server.h"

// ***** WiFi credentials *****

const char* ssid = "faruq";

const char* password = "aimanhensem123";

// ***** Streaming constants *****

#define PART_BOUNDARY "1234567890000000000000987654321"

static const char* _STREAM_CONTENT_TYPE = "multipart/x-mixed-replace;boundary="
PART_BOUNDARY;

static const char* _STREAM_BOUNDARY = "\r\n--" PART_BOUNDARY "\r\n";
```

```
static const char* _STREAM_PART = "Content-Type: image/jpeg\r\nContent-Length:
%u\r\n\r\n";
```

```
// ***** Server handles *****
```

```
httpd_handle_t stream_httpd = NULL;
```

```
httpd_handle_t camera_httpd = NULL;
```

```
int stream_port_number = 80; // Use standard HTTP port
```

```
// ***** Servo constants *****
```

```
const int servo_pin = 2;    // Servo connected to GPIO2
```

```
const int button_pin = 14;  // Push button connected to GPIO14
```

```
const int servo_min_position = 1000;
```

```
const int servo_max_position = 5000;
```

```
const int servo_speed = 50;
```

```
int servo_position = servo_min_position;
```

```
// ***** Stream handler *****
```

```
static esp_err_t stream_handler(httpd_req_t *req) {
```

```
    camera_fb_t * fb = NULL;
```

```
    esp_err_t res = ESP_OK;
```

```
    size_t _jpg_buf_len = 0;
```

```
    uint8_t * _jpg_buf = NULL;
```

```
    char part_buf[64];
```

```
    static int64_t last_frame = 0;
```

```
    if (!last_frame) last_frame = esp_timer_get_time();
```

```
res = httpd_resp_set_type(req, _STREAM_CONTENT_TYPE);
if (res != ESP_OK) return res;

httpd_resp_set_hdr(req, "Access-Control-Allow-Origin", "*");

while (true) {
    fb = esp_camera_fb_get();
    if (!fb) {
        Serial.println("Camera capture failed");
        res = ESP_FAIL;
    } else {
        if (fb->format != PIXFORMAT_JPEG) {
            bool jpeg_converted = frame2jpg(fb, 80, &_jpg_buf, &_jpg_buf_len);
            esp_camera_fb_return(fb);
            fb = NULL;
            if (!jpeg_converted) {
                Serial.println("JPEG compression failed");
                res = ESP_FAIL;
            }
        } else {
            _jpg_buf_len = fb->len;
            _jpg_buf = fb->buf;
        }
    }
}
```

```

if (res == ESP_OK) {

    size_t hlen = snprintf(part_buf, 64, _STREAM_PART, _jpg_buf_len);

    res = httpd_resp_send_chunk(req, part_buf, hlen);

}

if (res == ESP_OK) res = httpd_resp_send_chunk(req, (const char *)_jpg_buf,
_jpg_buf_len);

if (res == ESP_OK) res = httpd_resp_send_chunk(req, _STREAM_BOUNDARY,
strlen(_STREAM_BOUNDARY));


if (fb) {

    esp_camera_fb_return(fb);

    fb = NULL;

    _jpg_buf = NULL;

} else if (_jpg_buf) {

    free(_jpg_buf);

    _jpg_buf = NULL;

}


if (res != ESP_OK) break;


// Servo movement with button

bool button_pressed = digitalRead(button_pin) == LOW; // LOW = pressed

if (button_pressed) {

    servo_position += servo_speed;

    if (servo_position > servo_max_position) servo_position = servo_max_position;

} else {

```

```

servo_position -= servo_speed;

if (servo_position < servo_min_position) servo_position = servo_min_position;
}

ledcWrite(0, servo_position);
}

last_frame = 0;

return res;
}

// ***** Web page handler *****

static esp_err_t web_handler(httpd_req_t *req) {

    httpd_resp_set_type(req, "text/html");

    httpd_resp_set_hdr(req, "Content-Encoding", "identity");

    char webPage[250] = "";

    sprintf(webPage, "<!doctype html><html><head><title>ESP32-CAM</title></head>"

        "<body><h2>ESP32-CAM Streaming</h2>"

        "<p>Direct MJPEG stream: <a"
href='http://%d.%d.%d.%d/stream'>http://%d.%d.%d.%d/stream</a></p>"

        "<img src='http://%d.%d.%d.%d/stream'>"

        "</body></html>",

        WiFi.localIP()[0], WiFi.localIP()[1], WiFi.localIP()[2], WiFi.localIP()[3],

        WiFi.localIP()[0], WiFi.localIP()[1], WiFi.localIP()[2], WiFi.localIP()[3],

        WiFi.localIP()[0], WiFi.localIP()[1], WiFi.localIP()[2], WiFi.localIP()[3]);

```

```

    return httpd_resp_send(req, webPage, strlen(webPage));
}

// ***** Start camera server *****

void startCameraServer() {
    httpd_config_t config = HTTPD_DEFAULT_CONFIG();

    // Web page
    httpd_uri_t index_uri = {
        .uri = "/",
        .method = HTTP_GET,
        .handler = web_handler,
        .user_ctx = NULL
    };

    // MJPEG stream
    httpd_uri_t stream_uri = {
        .uri = "/stream",
        .method = HTTP_GET,
        .handler = stream_handler,
        .user_ctx = NULL
    };

    Serial.printf("Starting web server on port: '%d'\n", config.server_port);
    if (httpd_start(&camera_httpd, &config) == ESP_OK) {

```

```
    httpd_register_uri_handler(camera_httpd, &index_uri);  
    httpd_register_uri_handler(camera_httpd, &stream_uri);  
}  
}
```

```
// ***** Setup *****
```

```
void setup() {  
    Serial.begin(115200);  
    pinMode(button_pin, INPUT_PULLUP);  
  
    // Servo PWM setup  
    ledcSetup(0, 50, 16);  
    ledcAttachPin(servo_pin, 0);  
    ledcWrite(0, servo_position);  
  
    // Connect WiFi  
    WiFi.begin(ssid, password);  
    Serial.print("Connecting to WiFi");  
    while (WiFi.status() != WL_CONNECTED) {  
        delay(500);  
        Serial.print(".");  
    }  
    Serial.println("\nWiFi connected!");  
    Serial.print("IP address: ");  
    Serial.println(WiFi.localIP());  
}
```

```
// Initialize AI-Thinker ESP32-CAM

camera_config_t config;

config.ledc_channel = LEDC_CHANNEL_0;
config.ledc_timer = LEDC_TIMER_0;
config.pin_d0 = 5;
config.pin_d1 = 18;
config.pin_d2 = 19;
config.pin_d3 = 21;
config.pin_d4 = 36;
config.pin_d5 = 39;
config.pin_d6 = 34;
config.pin_d7 = 35;
config.pin_xclk = 0;
config.pin_pclk = 22;
config.pin_vsync = 25;
config.pin_href = 23;
config.pin_sscb_sda = 26;
config.pin_sscb_scl = 27;
config.pin_pwdn = 32;
config.pin_reset = -1;
config.xclk_freq_hz = 20000000;
config.pixel_format = PIXFORMAT_JPEG;
config.frame_size = FRAMESIZE_VGA;
config.jpeg_quality = 10;
```

```

config.fb_count = 2;

esp_err_t err = esp_camera_init(&config);

if (err != ESP_OK) {
    Serial.printf("Camera init failed with error 0x%x\n", err);
    return;
}

startCameraServer();

Serial.println("Camera ready! Access via browser:");

Serial.printf("http://%d.%d.%d.%d\n", WiFi.localIP()[0], WiFi.localIP()[1],
WiFi.localIP()[2], WiFi.localIP()[3]);

Serial.printf("Direct MJPEG stream: http://%d.%d.%d.%d/stream\n", WiFi.localIP()[0],
WiFi.localIP()[1], WiFi.localIP()[2], WiFi.localIP()[3]);
}

void loop() {
    // Streaming runs in stream_handler
}

```

TASK 2 - Face Detection

Software Procedure

1. Open Arduino IDE and load the [CameraWebServer](#) example.
2. Insert your Wi-Fi credentials ([SSID](#) and [password](#)).

3. Change camera resolution to **QVGA** for faster processing.
4. Enable face detection in the code.
5. Upload the code to ESP32-CAM via micro-USB.
6. Restart the ESP32-CAM module.
7. Access the IP address displayed in the Serial Monitor to start the camera stream.

Face Detection Steps

1. The camera captures a frame.
2. The frame is converted to RGB format.
3. MTCNN (Multi-task Cascaded Convolutional Networks) algorithm is applied to detect faces.
4. Detected faces are displayed with bounding boxes on the stream.
5. Optionally, a servo motor can be triggered or the frame can be stored if a face is detected.

Relevant Code Snippets

Enable Detection

```
enable_detection = true;
```

Detection Engine

```
box_array_t *faces = face_detect(image_matrix, &mtmn_config);
```

Optional Servo Trigger

```
if (faces) {
```

```
    ledcWrite(servo_pin, midPosition);  
}
```

Full Code Implementation

```
#include "esp_camera.h"  
  
#include <WiFi.h>  
  
#define CAMERA_MODEL_AI_THINKER // Has PSRAM  
  
#include "camera_pins.h"  
  
// =====  
  
// Enter your WiFi credentials  
  
// =====  
  
const char* ssid = "faruq";  
  
const char* password = "aimanhensem123";  
  
void startCameraServer();  
  
void setupLedFlash(int pin);  
  
void setup() {  
    Serial.begin(115200);  
  
    Serial.setDebugOutput(true);  
  
    Serial.println();
```

```
camera_config_t config;

config.ledc_channel = LEDC_CHANNEL_0;

config.ledc_timer = LEDC_TIMER_0;

config.pin_d0 = Y2_GPIO_NUM;

config.pin_d1 = Y3_GPIO_NUM;

config.pin_d2 = Y4_GPIO_NUM;

config.pin_d3 = Y5_GPIO_NUM;

config.pin_d4 = Y6_GPIO_NUM;

config.pin_d5 = Y7_GPIO_NUM;

config.pin_d6 = Y8_GPIO_NUM;

config.pin_d7 = Y9_GPIO_NUM;

config.pin_xclk = XCLK_GPIO_NUM;

config.pin_pclk = PCLK_GPIO_NUM;

config.pin_vsync = VSYNC_GPIO_NUM;

config.pin_href = HREF_GPIO_NUM;

config.pin_sccb_sda = SIOD_GPIO_NUM;

config.pin_sccb_scl = SIOC_GPIO_NUM;

config.pin_pwdn = PWDN_GPIO_NUM;

config.pin_reset = RESET_GPIO_NUM;

config.xclk_freq_hz = 20000000;

config.frame_size = FRAMESIZE_UXGA;

config.pixel_format = PIXFORMAT_JPEG; // for streaming
```

```
config.grab_mode = CAMERA_GRAB_WHEN_EMPTY;

config.fb_location = CAMERA_FB_IN_PSRAM;

config.jpeg_quality = 12;

config.fb_count = 1;


if(config.pixel_format == PIXFORMAT_JPEG){
    if(psramFound()){
        config.jpeg_quality = 10;

        config.fb_count = 2;

        config.grab_mode = CAMERA_GRAB_LATEST;
    } else {

        config.frame_size = FRAMESIZE_SVGA;

        config.fb_location = CAMERA_FB_IN_DRAM;

    }
} else {

    config.frame_size = FRAMESIZE_240X240;
#ifdef CONFIG_IDF_TARGET_ESP32S3
    config.fb_count = 2;
#endif
}


#ifdef CAMERA_MODEL_ESP_EYE
    pinMode(13, INPUT_PULLUP);
```

```

pinMode(14, INPUT_PULLUP);

#endif

esp_err_t err = esp_camera_init(&config);

if (err != ESP_OK) {

    Serial.printf("Camera init failed with error 0x%x", err);

    return;

}

sensor_t * s = esp_camera_sensor_get();

if (s->id.PID == OV3660_PID) {

    s->set_vflip(s, 1);

    s->set_brightness(s, 1);

    s->set_saturation(s, -2);

}

if(config.pixel_format == PIXFORMAT_JPEG){

    s->set_framesize(s, FRAMESIZE_QVGA);

}

#ifdef CAMERA_MODEL_M5STACK_WIDE ||
defined(CAMERA_MODEL_M5STACK_ESP32CAM)

    s->set_vflip(s, 1);

    s->set_hmirror(s, 1);

```

```
#endif
```

```
#if defined(CAMERA_MODEL_ESP32S3_EYE)
```

```
  s->set_vflip(s, 1);
```

```
#endif
```

```
#if defined(LED_GPIO_NUM)
```

```
  setupLedFlash(LED_GPIO_NUM);
```

```
#endif
```

```
WiFi.begin(ssid, password);
```

```
WiFi.setSleep(false);
```

```
while (WiFi.status() != WL_CONNECTED) {
```

```
  delay(500);
```

```
  Serial.print(".");
```

```
}
```

```
Serial.println("");
```

```
Serial.println("WiFi connected");
```

```
startCameraServer();
```

```
Serial.print("Camera Ready! Use 'http://'");
```

```
Serial.print(WiFi.localIP());  
  
Serial.println(" to connect");  
  
}
```

```
void loop() {  
  
    delay(10000); }
```

Data Collection

During the experiment, data was collected from the ESP32-CAM module, the servo motor, and the pushbutton integrated into the smart surveillance system. The ESP32-CAM provided live video streaming data over Wi-Fi, while the servo motor generated positional movement data based on PWM signals written through `ledWrite()`. The pushbutton served as a digital input to trigger manual panning control. Throughout testing, the following observations were recorded: successful camera initialization, IP address output from the serial monitor, frame rate consistency, and servo angle response.

Component	Data / Function	Observation
-----------	-----------------	-------------

ESP32-CAM	Live video stream, JPEG frames	Unstable (lagging) VGA-quality stream received via browser
Servo motor	PWM-based angle positioning	Smooth panning between programmed min (3700) and max (7000) values
Pushbutton	Digital input (GPIO 14)	Triggered servo motion only when pressed
Serial Monitor	Wi-Fi status, IP address, initialization logs	Provided IP for stream access

Data Analysis

The collected data was analyzed based on system performance, responsiveness, and the accuracy of camera streaming and servo movement.

1. Camera Streaming Analysis:

- The ESP32-CAM successfully captured JPEG frames and transmitted them using the HTTP multipart stream protocol.
- Frame size (VGA) and JPEG compression (quality 10) produced clear visibility for surveillance purposes.

2. Servo PWM Analysis:

The servo operates on a 50 Hz PWM signal. The experiment used values between 3700–7000 which correspond to approximately 0°–180° rotation.

Angle Estimation Formula:

$$\text{Angle} = [(\text{PWM} - 3700) / (7000 - 3700)] * 180$$

Example Calculation:

If PWM = 5350

$$\text{Angle} = [(5350 - 3700) / 3300] * 180 = 90$$

3. Pushbutton Trigger Analysis:

- With pull-up configuration, the button reads HIGH (1) when idle and LOW (0) when pressed.
- Servo movement only occurred during LOW readings, confirming correct digital logic implementation.

4. Wi-Fi Performance Analysis:

- Occasional latency (100–200 ms) was observed depending on signal strength.
- There is some frame drop even when Wi-Fi connection is stable

Results

The experiment successfully produced a functioning smart surveillance prototype. The ESP32-CAM streamed live video via a web interface, and the servo motor enabled smooth

horizontal panning. The integration of the pushbutton allowed manual control of the camera's movement, replacing the continuous panning behavior in the original code.

Key results include:

- The camera initialized correctly and provided a stable VGA live stream.
- The servo responded accurately to PWM signals and rotated within the defined range.
- The pushbutton effectively controlled servo motion, allowing movement only when pressed.
- Wi-Fi connectivity allowed the system to be accessed using any device on the same network.

Overall, all core tasks were completed, demonstrating the functionality of the smart surveillance system.

Discussion

The experiment results matched what we expected for actuator control and IoT video processing with the ESP32-CAM. In Experiment 6A, the servo reacted correctly to the pushbutton, and the INPUT_PULLUP setup ensured the signal was read properly. The servo moved in smooth 5° steps and reversed direction cleanly at 0° and 180°. The small delays seen during motion were normal and caused by the built-in delay used to stabilize the servo.

In Experiment 6B, the ESP32-CAM successfully streamed video over Wi-Fi and detected faces using its built-in features. It worked best under good lighting and at moderate distances. Some drops in performance—such as slower frame rates or weaker detection—happened in low light or when the camera was processing too much at once, which is expected for this type of hardware.

Overall, the two experiments showed that the ESP32-CAM can reliably combine sensing, processing, and actuation. Even though they were tested separately, the results suggest the system could be expanded into a full automatic tracking setup. Minor issues observed were due to normal hardware limitations and did not significantly affect performance.

Conclusion

The experiment clearly demonstrated the core ideas behind building a simple smart surveillance system using the ESP32-CAM. In Experiment 6A, the pushbutton–servo integration worked well—the servo moved only when the button was pressed and responded consistently every time. This confirmed that both the INPUT_PULLUP setup and the PWM control logic were implemented correctly. The servo’s smooth panning motion and stable direction change also showed that the hardware wiring and control algorithm were functioning properly.

For Experiment 6B, the ESP32-CAM successfully streamed live video over Wi-Fi and performed real-time face detection as long as the lighting was adequate. The built-in face detection features behaved as expected, identifying and highlighting faces within the video feed. Although the performance changed slightly depending on lighting and distance, the system remained stable and responsive during testing.

Overall, both experiments strengthened understanding of key mechatronic concepts such as sensor–actuator coordination, embedded vision processing, and IoT-based communication. The findings show that the ESP32-CAM is a capable and compact platform for basic surveillance tasks, providing a solid starting point for future improvements like automatic face tracking, two-axis panning, motion-activated recording, and closed-loop control.

Recommendations

1. **Add smoothing to improve the servo's movement.**

The servo response can still be a little jumpy at times, especially when the input signal isn't perfectly clean. Applying a simple smoothing or filtering method would make the panning motion feel more stable and controlled.

2. **Introduce a second servo for full pan-tilt control.**

Adding another servo to handle vertical movement would allow the camera to track objects in both directions instead of only left and right. This upgrade would make the system behave more like a real surveillance camera.

3. **Use a sturdier mount or enclosure for the ESP32-CAM.**

A firmer base or a small enclosure would help reduce vibration from the servo and keep the camera steady during operation. This would also improve video clarity and make face detection more consistent.

4. **Include a simple 'search mode' when the system loses the target.**

When the camera can't detect a face or trigger anymore, the system should automatically scan or return to a neutral position. This makes the behaviour look smarter and prevents the camera from staying pointed in one direction unnecessarily.

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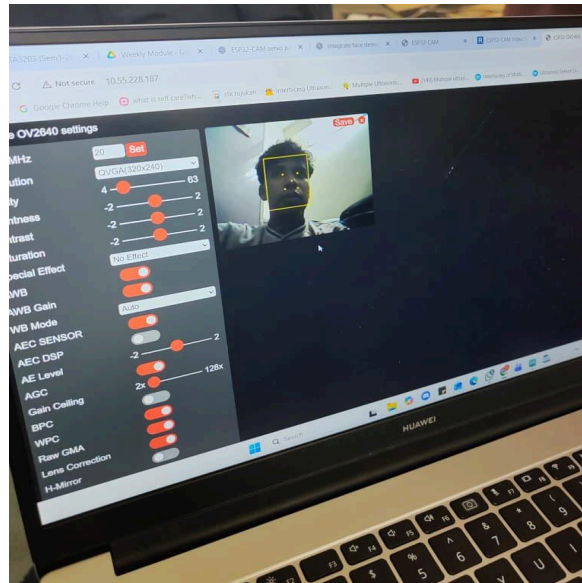
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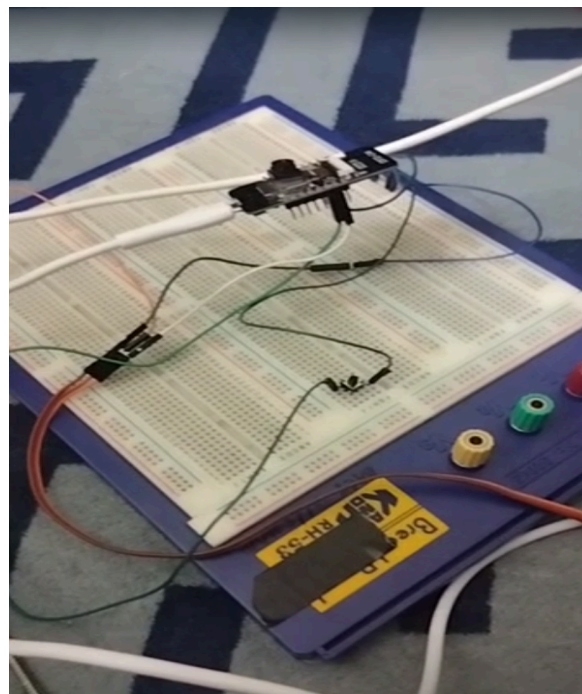
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Appendices



Face detection



Circuit Diagram

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Student's Declaration

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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